Tarea 09

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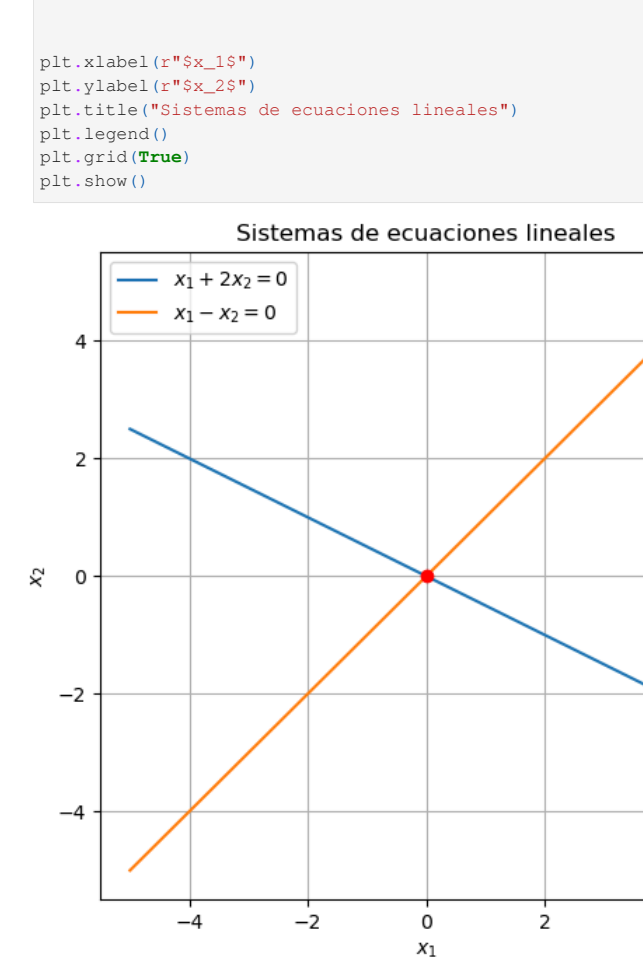
1)Para cada uno de los siguientes sistemas lineales, obtenga, de ser posible, una solución con métodos gráficos. Explique los resultados desde un punto de vista geométrico.

a.
$x_1 + 2x_2 = 0$
$x_1 - x_2$

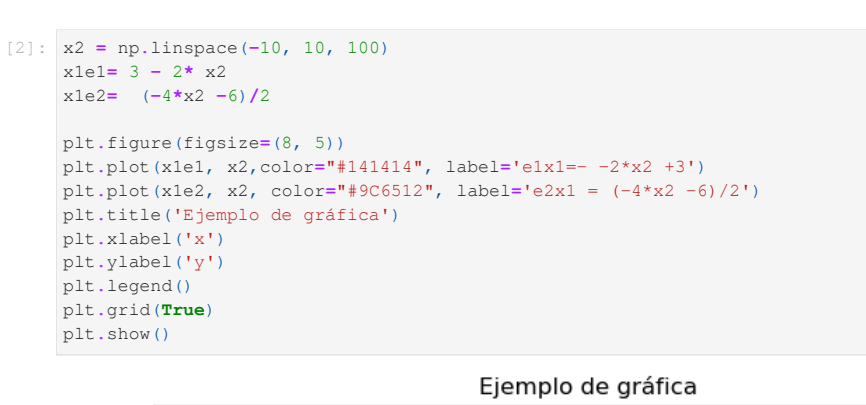
In [1]:
```
import numpy as np
import matplotlib.pyplot as plt


x1 = np.linspace(-5, 5, 400)

x2_eq1 = -0.5 * x1
x2_eq2 = x1

plt.figure(figsize=(6,6))
plt.plot(x1, x2_eq1, label=r"$x_1 + 2x_2 = 0$")
plt.plot(x1, x2_eq2, label=r"$x_1 - x_2 = 0$")

# Intersección

plt.scatter(0, 0, color = "red", zorder=5)


plt.xlabel(r"$x_1$")
plt.ylabel(r"$x_2$")
plt.title("Sistemas de ecuaciones lineales")
plt.legend()
plt.grid(True)
plt.show()
```



El sistema tiene una única solución, ya que las rectas se intersecan en un único punto el cuál representa la solución

b.
$x_1 + 2x_2 = 3$
$-2x_1 - 4x_2 = 6$

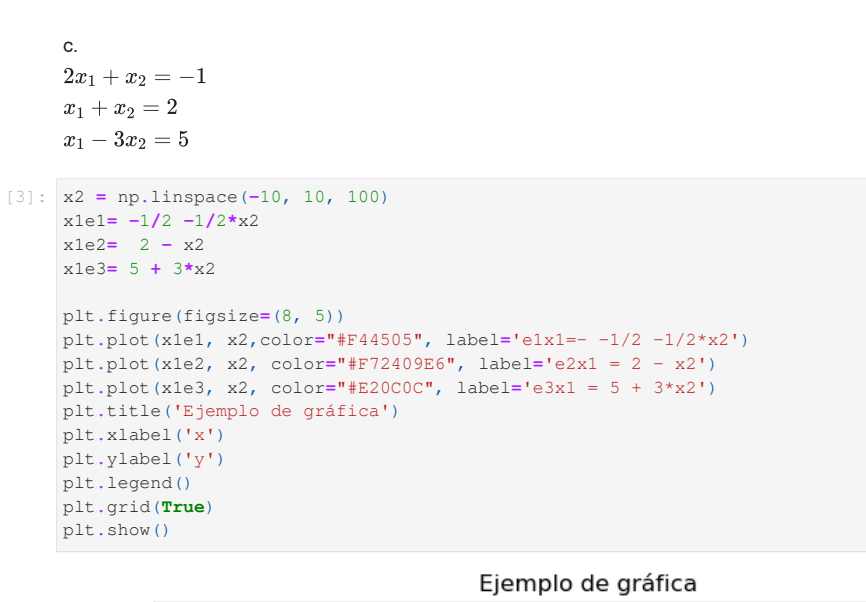
In [2]:
```
x2 = np.linspace(-10, 10, 100)
x1e1= 3 - 2* x2
x1e2=  (-4*x2 -6)/2

plt.figure(figsize=(8, 5))
plt.plot(x1e1, x2,color="#141414", label='e1x1=- -2*x2 +3')
plt.plot(x1e2, x2, color="#9C6512", label='e2x1 = (-4*x2 -6)/2')
plt.title('Ejemplo de gráfica')
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.show()
```



Ambas rectan no intersecan en ningún punto, son paralelas y por lo tanto no hay raíz

c.
$2x_1 + x_2 = -1$
$x_1 + x_2 = 2$
$x_1 - 3x_2 = 5$

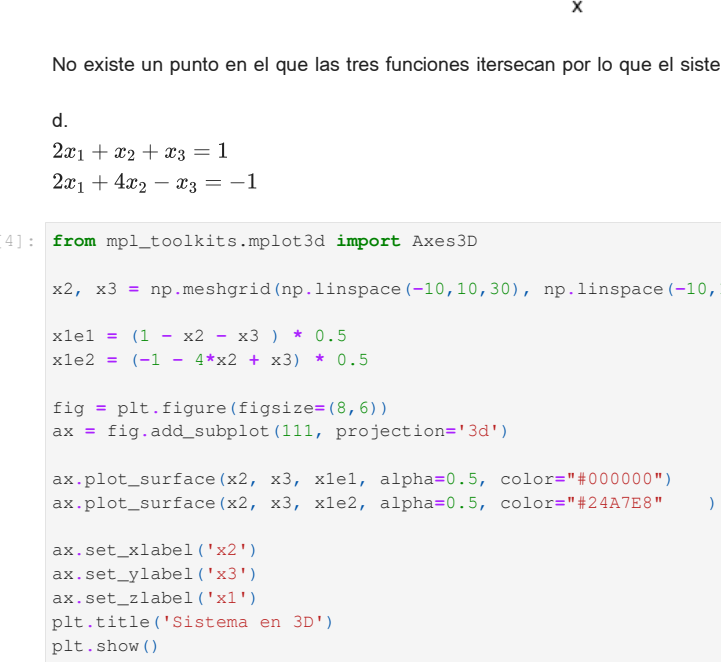
In [3]:
```
x2 = np.linspace(-10, 10, 100)
x1e1= -1/2 -1/2*x2
x1e2=  2 - x2
x1e3= 5 + 3*x2

plt.figure(figsize=(8, 5))
plt.plot(x1e1, x2,color="#F44505", label='e1x1=- -1/2 -1/2*x2')
plt.plot(x1e2, x2, color="#F72409E6", label='e2x1 = 2 - x2')
plt.plot(x1e3, x2, color="#E20C0C", label='e3x1 = 5 + 3*x2')
plt.title('Ejemplo de gráfica')
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.show()
```



No existe un punto en el que las tres funciones itersecan por lo que el sistema no tiene solución,

d.
$2x_1 + x_2 + x_3 = 1$
$2x_1 + 4x_2 - x_3 = -1$

In [4]:
```
from mpl_toolkits.mplot3d import Axes3D

x2, x3 = np.meshgrid(np.linspace(-10,10,30), np.linspace(-10,10,30))

x1e1 = (1 - x2 - x3 ) * 0.5
x1e2 = (-1 - 4*x2 + x3) * 0.5

fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.plot_surface(x2, x3, x1e1, alpha=0.5, color="#000000")
ax.plot_surface(x2, x3, x1e2, alpha=0.5, color="#24A7E8"    )

ax.set_xlabel('x2')
ax.set_ylabel('x3')
ax.set_zlabel('x1')
plt.title('Sistema en 3D')
plt.show()
```



2) Utilice la eliminación gaussiana con sustitución hacia atrás y aritmética de redondeo de dos dígitos para resolver los siguientes sistemas lineales. No reordene las ecuaciones. (La solución exacta para cada sistema es $x1$ =−1

In [5]:
```
import numpy as np
import logging
from sys import stdout
from datetime import datetime

logging.basicConfig(
    level=logging.INFO,
    format="[%(asctime)s][%(levelname)s] %(message)s",
    stream=stdout,
    datefmt="%m-%d %H:%M:%S",
)
logging.info(datetime.now())

def r(x):
    return np.round(x, 2)

def eliminacion_gaussiana(A: np.ndarray | list[list[float | int]]) -> np.ndarray:
    """Resuelve un sistema de ecuaciones lineales mediante el método de eliminación gaussiana.

    ## Parameters

    ``A``: matriz aumentada del sistema de ecuaciones lineales. Debe ser de tamaño n-by-(n+1), donde n es el número de incógnitas.

    ## Return

    ``solucion``: vector con la solución del sistema de ecuaciones lineales.

    """
    if not isinstance(A, np.ndarray):
        logging.debug("Convirtiendo A a numpy array.")
        A = np.array(A)
    assert A.shape[0] == A.shape[1] - 1, "La matriz A debe ser de tamaño n-by-(n+1)."
    n = A.shape[0]

    for i in range(0, n - 1 ): #loop por columna

        # ---encontrar pivote
        p = None #default, first element
        for pi in range(i, n):
            if A[pi, i] == 0:
                # must be nonzero
                continue

            if p is None:
                # first nonzero element
                p = pi
                continue

            if abs(A[pi, i] < abs(A[p,i])):
                p = pi
        if p is None:
            # no pivot found.
            raise ValueError("No existe solución única.")

        if p != i:
            # swap rows
            logging.debug(f"Intercambiando filas {i} y {p}")
            _aux = A[i, :].copy()
            A[i, :] = A[p, :].copy()
            A[p, :] = _aux

        # --- Eliminación: loop por fila
        for j in range(i + 1, n):
            m = r( A[j, i] / A[i, i])
            A[j, i:] = r(A[j, i:] - r(m * A[i, i:]))

        logging.info(f"\n{A}")

    if A[n - 1, n - 1] == 0:
        raise ValueError("No existe solución única.")

        print(f"\n{A}")
    # --- Sustitución hacia atrás
    solucion = np.zeros(n)
    solucion[n - 1] =r( A[n - 1, n] / A[n - 1, n - 1])

    for i in range(n - 2, -1, -1):
        suma = 0
        for j in range(i + 1, n):
            suma += r(A[i, j] * solucion[j])
        solucion[i] = r((A[i, n] - suma) / A[i, i])

    return solucion
```

```
[07-08 21:06:35][INFO] 2026-07-08 21:06:35.654845
```

a.
$-x_1 + 4x_2 + x_3 = 8$

$\frac{5}{3}x_1 + \frac{2}{3}x_2 + \frac{2}{3}x_3$

$2x_1 + x_2 + 4x_3 = 11$

In [6]:
```
Aa = [
    [-1., 4, 1, 8],
    [5/3, 2/3, 2/3, 1],
    [2, 1, 4, 11]
]

eliminacion_gaussiana(Aa)
```

```
[07-08 21:06:35][INFO]
[[-1.    4.    1.    8.  ]
 [-0.    7.35  2.34 14.36]
 [ 0.    9.    6.   27.  ]]
[07-08 21:06:35][INFO]
[[-1.    4.    1.    8.  ]
 [-0.    7.35  2.34 14.36]
 [ 0.    0.03  3.15  9.48]]
```

Out[6]:
```
array([-0.99,  1.  ,  3.01])
```

b.
$4x_1 + 2x_2 - x_3 = -5$

$\frac{1}{9}x_1 + \frac{1}{9}x_2 - \frac{1}{3}x_3 = -1$

$x_1 + 4x_2 + 2x_3$

In [7]:
```
Ab = [
    [4., 2,-1,-5],
    [1/9,1/9,-1/3,-1],
    [1,4,2,9]
]
eliminacion_gaussiana(Ab)
```

```
[07-08 21:06:35][INFO]
[[ 0.11111111  0.11111111 -0.33333333 -1.        ]
 [ 0.         -2.         11.         31.        ]
 [ 0.          3.          5.         18.        ]]
[07-08 21:06:35][INFO]
[[ 0.11111111  0.11111111 -0.33333333 -1.        ]
 [ 0.         -2.         11.         31.        ]
 [ 0.          0.         21.5        64.5       ]]
```

Out[7]:
```
array([-0.99,  1.  ,  3.  ])
```

3) Utilice el algoritmo de eliminación gaussiana para resolver, de ser posible, los siguientes sistemas lineales, y determine si se necesitan intercambios de fila

In [8]:
```
import numpy as np
import logging
from sys import stdout
from datetime import datetime

intercambio=0
logging.basicConfig(
    level=logging.INFO,
    format="[%(asctime)s][%(levelname)s] %(message)s",
    stream=stdout,
    datefmt="%m-%d %H:%M:%S",
)
logging.info(datetime.now())

def eliminacion_gaussiana(A: np.ndarray | list[list[float | int]]) -> np.ndarray:
    """Resuelve un sistema de ecuaciones lineales mediante el método de eliminación gaussiana.

    ## Parameters

    ``A``: matriz aumentada del sistema de ecuaciones lineales. Debe ser de tamaño n-by-(n+1), donde n es el número de incógnitas.

    ## Return

    ``solucion``: vector con la solución del sistema de ecuaciones lineales.

    """
    if not isinstance(A, np.ndarray):
        logging.debug("Convirtiendo A a numpy array.")
        A = np.array(A)
    assert A.shape[0] == A.shape[1] - 1, "La matriz A debe ser de tamaño n-by-(n+1)."
    n = A.shape[0]

    for i in range(0, n - 1):  # loop por columna

        # --- encontrar pivote
        p = None  # default, first element
        for pi in range(i, n):
            if A[pi, i] == 0:
                # must be nonzero
                continue

            if p is None:
                # first nonzero element
                p = pi
                continue

            if abs(A[pi, i]) < abs(A[p, i]):
                p = pi

        if p is None:
            # no pivot found.
            raise ValueError("No existe solución única.")

        if p != i:
            # swap rows
            logging.debug(f"Intercambiando filas {i} y {p}")
            _aux = A[i, :].copy()
            A[i, :] = A[p, :].copy()
            A[p, :] = _aux
            global intercambio
            intercambio+=1

        # --- Eliminación: loop por fila
        for j in range(i + 1, n):
            m = A[j, i] / A[i, i]
            A[j, i:] = A[j, i:] - m * A[i, i:]

        logging.info(f"\n{A}")

    if A[n - 1, n - 1] == 0:
        print( ValueError("No existe solución única."))
        return

        print(f"\n{A}")
    # --- Sustitución hacia atrás
    solucion = np.zeros(n)
    solucion[n - 1] = A[n - 1, n] / A[n - 1, n - 1]

    for i in range(n - 2, -1, -1):
        suma = 0
        for j in range(i + 1, n):
            suma += A[i, j] * solucion[j]
        solucion[i] = (A[i, n] - suma) / A[i, i]

    print("Número de intercambios de filas:", intercambio)
    return solucion
```

```
[07-08 21:06:35][INFO] 2026-07-08 21:06:35.709723
```

a.
$x_1 - x_2 + 3x_3 = 2$
$3x_1 - 3x_2 + x_3 = -1$
$x_1 + x_2 = 3$

In [9]:
```
A_a = [
    [1.,-1,3,2],
    [3,-3,1,-1],
    [1,1,0,3]
]

eliminacion_gaussiana(A_a)
```

```
[07-08 21:06:35][INFO]
[[ 1. -1.  3.  2.]
 [ 0.  0. -8. -7.]
 [ 0.  2. -3.  1.]]
[07-08 21:06:35][INFO]
[[ 1. -1.  3.  2.]
 [ 0.  2. -3.  1.]
 [ 0.  0. -8. -7.]]
Número de intercambios de filas: 1
```

Out[9]:
```
array([1.1875, 1.8125, 0.875 ])
```

b.
$2x_1 - 1.5x_2 + 3x_3 = 1$
$-x_1 + 2x_3 = 3$
$4x_1 - 4.5x_2 + 5x_3 = 1$

In [10]:
```
A_b = [[2., -1/5, 3,1], [-1,0 , 2, 3], [4,-4.5, 5, 1]]

eliminacion_gaussiana(A_b)
```

```
[07-08 21:06:35][INFO]
[[-1.    0.    2.    3.  ]
 [ 0.   -0.2   7.    7.  ]
 [ 0.   -4.5 13.   13.  ]]
[07-08 21:06:35][INFO]
[[  -1.      0.      2.      3. ]
 [   0.     -0.2     7.      7. ]
 [   0.      0.  -144.5 -144.5]]
Número de intercambios de filas: 2
```

Out[10]:
```
array([-1., -0.,  1.])
```

c.
$2x_1 = 3$
$x_1 + 1.5x_2 = 4.5$
$-3x_2 + 0.5x_3 = -6.6$
$2x_1 - 2x_2 + x_3 + x_4 = 0.8$

In [11]:
```
A_c = [
    [2.,0,0,0,3],
    [1,1.5,0,0,4.5],
    [0,-3,0.5,0,-6.6],
    [2,-2,1,1,0.8]
]
eliminacion_gaussiana(A_c)
```

```
[07-08 21:06:35][INFO]
[[ 1.   1.5  0.   0.   4.5]
 [ 0.  -3.   0.   0.  -6. ]
 [ 0.  -3.   0.5  0.  -6.6]
 [ 0.  -5.   1.   1.  -8.2]]
[07-08 21:06:35][INFO]
[[ 1.   1.5  0.   0.   4.5]
 [ 0.  -3.   0.   0.  -6. ]
 [ 0.   0.   0.5  0.  -0.6]
 [ 0.   0.   1.   1.   1.8]]
[07-08 21:06:35][INFO]
[[ 1.   1.5  0.   0.   4.5]
 [ 0.  -3.   0.   0.  -6. ]
 [ 0.   0.   0.5  0.  -0.6]
 [ 0.   0.   0.   1.   3. ]]
Número de intercambios de filas: 3
```

Out[11]:
```
array([ 1.5,  2. , -1.2,  3. ])
```

d.
$x_1$ + x_2 + x_4 = 2
$2x_1 + x_2 - x_3 + x_4 = 1$
$4x_1 - x_2 - 2x_3 + 2x_4 = 0$
$3x_1 - x_2 - x_3 + 2x_4 = -3$

In [12]:
```
A_d = [
    [1,1,0,1,2],
    [2,1,-1,1,1],
    [4,-1,-2,2,0],
    [3,-1,-1,2,-3]
]

eliminacion_gaussiana(A_d)
```

```
[07-08 21:06:35][INFO]
[[ 1  1  0  1  2]
 [ 0 -1 -1 -1 -3]
 [ 0 -5 -2 -2 -8]
 [ 0 -4 -1 -1 -9]]
[07-08 21:06:35][INFO]
[[ 1  1  0  1  2]
 [ 0 -1 -1 -1 -3]
 [ 0  0  3  3  7]
 [ 0  0  3  3  3]]
[07-08 21:06:35][INFO]
[[ 1  1  0  1  2]
 [ 0 -1 -1 -1 -3]
 [ 0  0  3  3  7]
 [ 0  0  0  0 -4]]
No existe solución única.
```

4) Use el algoritmo de eliminación gaussiana y la aritmética computacional de precisión de 32 bits para resolver los siguientes sistemas lineales.

In [13]:
```
import numpy as np
import logging
from sys import stdout
from datetime import datetime

intercambio=0
logging.basicConfig(
    level=logging.INFO,
    format="[%(asctime)s][%(levelname)s] %(message)s",
    stream=stdout,
    datefmt="%m-%d %H:%M:%S",
)
logging.info(datetime.now())

def eliminacion_gaussiana(A: np.ndarray | list[list[float | int]]) -> np.ndarray:
    """Resuelve un sistema de ecuaciones lineales mediante el método de eliminación gaussiana.

    ## Parameters

    ``A``: matriz aumentada del sistema de ecuaciones lineales. Debe ser de tamaño n-by-(n+1), donde n es el número de incógnitas.

    ## Return

    ``solucion``: vector con la solución del sistema de ecuaciones lineales.

    """
    if not isinstance(A, np.ndarray):
        logging.debug("Convirtiendo A a numpy array.")
        A = np.array(A , dtype=np.float32)
    assert A.shape[0] == A.shape[1] - 1, "La matriz A debe ser de tamaño n-by-(n+1)."
    n = A.shape[0]

    for i in range(0, n - 1):  # loop por columna

        # --- encontrar pivote
        p = None  # default, first element
        for pi in range(i, n):
            if A[pi, i] == 0:
                # must be nonzero
                continue

            if p is None:
                # first nonzero element
                p = pi
                continue

            if abs(A[pi, i]) < abs(A[p, i]):
                p = pi

        if p is None:
            # no pivot found.
            raise ValueError("No existe solución única.")

        if p != i:
            # swap rows
            logging.debug(f"Intercambiando filas {i} y {p}")
            _aux = A[i, :].copy()
            A[i, :] = A[p, :].copy()
            A[p, :] = _aux
            global intercambio
            intercambio+=1

        # --- Eliminación: loop por fila
        for j in range(i + 1, n):
            m = A[j, i] / A[i, i]
            A[j, i:] = A[j, i:] - m * A[i, i:]

        logging.info(f"\n{A}")

    if A[n - 1, n - 1] == 0:
        raise ValueError("No existe solución única.")

        print(f"\n{A}")
    # --- Sustitución hacia atrás
    solucion = np.zeros(n)
    solucion[n - 1] = A[n - 1, n] / A[n - 1, n - 1]

    for i in range(n - 2, -1, -1):
        suma = 0
        for j in range(i + 1, n):
            suma += A[i, j] * solucion[j]
        solucion[i] = (A[i, n] - suma) / A[i, i]

    print("Número de intercambios de filas:", intercambio)
    return solucion


Ab = [[1/4, 1/5, 1/6, 9], [1/3, 1/4, 1/5, 8], [1/2 ,1, 2, 8]]

eliminacion_gaussiana(Ab)
```

```
[07-08 21:06:35][INFO] 2026-07-08 21:06:35.794187
[07-08 21:06:35][INFO]
[[  0.25         0.2          0.16666667   9.        ]
 [  0.          -0.01666668  -0.02222224  -4.        ]
 [  0.           0.6          1.6666666  -10.        ]]
[07-08 21:06:35][INFO]
[[ 2.5000000e-01  2.0000000e-01  1.6666667e-01  9.0000000e+00]
 [ 0.0000000e+00 -1.6666681e-02 -2.2222236e-02 -4.0000000e+00]
 [ 0.0000000e+00 -5.9604645e-08  8.6666673e-01 -1.5399989e+02]]
Número de intercambios de filas: 0
```

Out[13]:
```
array([-227.07666725,  476.92263902, -177.69216919])
```

a.

$\frac{1}{4}x_1 + \frac{1}{5}x_2 + \frac{1}{6}x_3 = 9$

$\frac{1}{3}x_1 + \frac{1}{4}x_2 + \frac{1}{5}x_3 = 8$

$\frac{1}{2}x_1 + x_2 + 2x_3 = 8$

In [14]:
```
A_aa = [
    [1/4,1/5,1/6,9],
    [1/3,1/4,1/5,8],
    [1/2,1,2,8]
]

eliminacion_gaussiana(A_aa)
```

```
[07-08 21:06:35][INFO]
[[  0.25         0.2          0.16666667   9.        ]
 [  0.          -0.01666668  -0.02222224  -4.        ]
 [  0.           0.6          1.6666666  -10.        ]]
[07-08 21:06:35][INFO]
[[ 2.5000000e-01  2.0000000e-01  1.6666667e-01  9.0000000e+00]
 [ 0.0000000e+00 -1.6666681e-02 -2.2222236e-02 -4.0000000e+00]
 [ 0.0000000e+00 -5.9604645e-08  8.6666673e-01 -1.5399989e+02]]
Número de intercambios de filas: 0
```

Out[14]:
```
array([-227.07666725,  476.92263902, -177.69216919])
```

b.
$3.333x_1 + 15920x_2 - 10.333x_3 = 15913$

$2.222x_1 + 16.71x_2 + 9.612x_3 = 28.544$

$1.5611x_1 + 5.1791x_2 + 1.6852x_3 = 8,4254$

In [15]:
```
A_ab = [
    [3.333,15920,-10.333,15913],
    [2.222,16.71,9.612,28.544],
    [1.5611,5.1791,1.6852,8.4254]
]

eliminacion_gaussiana(A_ab)
```

```
[07-08 21:06:35][INFO]
[[ 1.5611000e+00  5.1791000e+00  1.6852000e+00  8.4253998e+00]
 [ 0.0000000e+00  9.3383007e+00  7.2133622e+00  1.6551662e+01]
 [ 0.0000000e+00  1.5908942e+04 -1.3930958e+01  1.5895012e+04]]
[07-08 21:06:35][INFO]
[[ 1.5611000e+00  5.1791000e+00  1.6852000e+00  8.4253998e+00]
 [ 0.0000000e+00  9.3383007e+00  7.2133622e+00  1.6551662e+01]
 [ 0.0000000e+00  0.0000000e+00 -1.2302779e+04 -1.2302777e+04]]
Número de intercambios de filas: 1
```

Out[15]:
```
array([0.99999975, 1.00000009, 0.99999982])
```

c.

$x_1 + \frac{1}{2}x_2 + \frac{1}{3}x_3 + \frac{1}{4}x_4 = \frac{1}{6}$

$\frac{1}{2}x_1 + \frac{1}{3}x_2 + \frac{1}{4}x_3 + \frac{1}{5}x_4 = \frac{1}{7}$

$\frac{1}{3}x_1 + \frac{1}{4}x_2 + \frac{1}{5}x_3 + \frac{1}{6}x_4 = \frac{1}{8}$

$\frac{1}{4}x_1 + \frac{1}{5}x_2 + \frac{1}{6}x_3 + \frac{1}{7}x_4 = \frac{1}{9}$

In [16]:
```
A_ac = [
    [1,1/2,1/3,1/4,1/6],
    [1/2,1/3,1/4,1/5,1/7],
    [1/3,1/4,1/5,1/6,1/8],
    [1/4,1/5,1/6,1/7,1/9]
]

eliminacion_gaussiana(A_ac)
```

```
[07-08 21:06:35][INFO]
[[ 0.25        0.2         0.16666667  0.14285715  0.11111111]
 [ 0.         -0.06666666 -0.08333334 -0.0857143  -0.07936507]
 [ 0.         -0.01666668 -0.02222224 -0.02380954 -0.02314815]
 [ 0.         -0.3        -0.33333334 -0.3214286  -0.2777778 ]]
[07-08 21:06:35][INFO]
[[ 2.5000000e-01  2.0000000e-01  1.6666667e-01  1.4285715e-01
   1.1111111e-01]
 [ 0.0000000e+00 -1.6666681e-02 -2.2222236e-02 -2.3809537e-02
  -2.3148149e-02]
 [ 0.0000000e+00  0.0000000e+00  5.5555180e-03  9.5237717e-03
   1.3227440e-02]
 [ 0.0000000e+00  2.9802322e-08  6.6666603e-02  1.0714275e-01
   1.3888860e-01]]
[07-08 21:06:35][INFO]
[[ 2.5000000e-01  2.0000000e-01  1.6666667e-01  1.4285715e-01
   1.1111111e-01]
 [ 0.0000000e+00 -1.6666681e-02 -2.2222236e-02 -2.3809537e-02
  -2.3148149e-02]
 [ 0.0000000e+00  0.0000000e+00  5.5555180e-03  9.5237717e-03
   1.3227440e-02]
 [ 0.0000000e+00  2.9802322e-08  0.0000000e+00 -7.1431771e-03
  -1.9841611e-02]]
Número de intercambios de filas: 3
```

Out[16]:
```
array([-0.03174083,  0.5951854 , -2.3808313 ,  2.77770114])
```

d.

$2x_1 + x_2 - x_3 + x_4 - 3x_5 = 7$

$x_1 + 2x_3 - x_4 + x_5 = 2$

$-2x_2 - x_3 + x_4 - x_5 = -5$

$3x_1 + x_2 - 4x_3 + 5x_5 = 6$

$x_1 - x_2 - x_3 - x_4 + x_5 = -3$

In [17]:
```
A_ad = [
    [2.,1,-1,1,-3,7],
    [1,0,2,-1,1,2],
    [0,-2,-1,1,-1,-5],
    [3,1,-4,0,5,6],
    [1,-1,-1,-1,1,-3]
]

eliminacion_gaussiana(A_ad)
```

```
[07-08 21:06:35][INFO]
[[  1.   0.   2.  -1.   1.   2.]
 [  0.   1.  -5.   3.  -5.   3.]
 [  0.  -2.  -1.   1.  -1.  -5.]
 [  0.   1. -10.   3.   2.   0.]
 [  0.  -1.  -3.   0.   0.  -5.]]
[07-08 21:06:35][INFO]
[[  1.   0.   2.  -1.   1.   2.]
 [  0.   1.  -5.   3.  -5.   3.]
 [  0.   0. -11.   7. -11.   1.]
 [  0.   0.  -5.   0.   7.  -3.]
 [  0.   0.  -8.   3.  -5.  -2.]]
[07-08 21:06:35][INFO]
[[  1.          0.          2.         -1.          1.          2.        ]
 [  0.          1.         -5.          3.         -5.          3.        ]
 [  0.          0.         -5.          0.          7.         -3.        ]
 [  0.          0.          0.          7.        -26.400002    7.6000004]
 [  0.          0.          0.          3.        -16.2         2.8000002]]
[07-08 21:06:35][INFO]
[[  1.          0.          2.         -1.          1.          2.        ]
 [  0.          1.         -5.          3.         -5.          3.        ]
 [  0.          0.         -5.          0.          7.         -3.        ]
 [  0.          0.          0.          3.        -16.2         2.8000002]
 [  0.          0.          0.          0.         11.399998    1.0666666]]
Número de intercambios de filas: 6
```

Out[17]:
```
array([1.88304105, 2.80701722, 0.73099417, 1.43859666, 0.09356727])
```

5)Dado el sistema lineal:

$$x_1 - x_2 + \alpha x_3 = -2$$
$$-x_1 + 2x_2 - \alpha x_3 = 3$$
$$\alpha x_1 + x_2 + x3 = 2$$

a. Encuentre el valor(es) de α para los que el sistema no tiene soluciones.
b. Encuentre el valor(es) de α para los que el sistema tiene un número infinito de soluciones.
c. Suponga que existe una única solución para una a determinada, encuentre la solución.

In [18]:
```
import sympy

def analyze_system_with_parameter(A_symbolic, b_symbolic, symbol):

    if not A_symbolic.is_square:
        print("Error: La matriz de coeficientes debe ser cuadrada.")
        return

    print("--- Análisis del Determinante ---")

    det_A = A_symbolic.det()
    print(f"La matriz de coeficientes A es:\n{A_symbolic}")
    print(f"\nEl determinante de A es: det(A) = {sympy.simplify(det_A)}")

    print("\nSe busca para qué valores del símbolo el determinante es cero.")
    critical_values = sympy.solve(det_A, symbol)
    print(f"El determinante es cero cuando {symbol} está en {critical_values}.")
    print("-" * 60)

    print("--- Análisis de los Casos Críticos (No hay solución única) ---")
    for val in critical_values:
        print(f"\nAnálisis para {symbol} = {val}:")

        A_sub = A_symbolic.subs(symbol, val)
        b_sub = b_symbolic.subs(symbol, val)
        augmented_matrix = A_sub.row_join(b_sub)
        print("Matriz aumentada para este valor:\n", augmented_matrix)

        rref_matrix, _ = augmented_matrix.rref()
        print("\nForma escalonada reducida por filas (RREF):\n", rref_matrix)

        inconsistent = any(all(elem == 0 for elem in row[:-1]) and row[-1] != 0 for row in rref_matrix.tolist())

        if inconsistent:
            print(f"\nConclusión (a): Para {symbol} = {val}, el sistema es inconsistente y NO TIENE SOLUCIONES.")
        else:
            print(f"\nConclusión (b): Para {symbol} = {val}, el sistema es consistente y tiene INFINITAS SOLUCIONES.")
    print("-" * 60)

    print("--- Análisis del Caso General (Solución única) ---")
    print(f"Existe una solución única siempre que {symbol} no sea uno de los valores críticos {critical_values}.")

    print("\nUsando la Regla de Cramer para encontrar la solución en términos del símbolo:")

    solutions = []
    for i in range(A_symbolic.cols):
        Ai = A_symbolic.copy()
        Ai[:, i] = b_symbolic

        det_Ai = Ai.det()
        xi = sympy.simplify(det_Ai / det_A)
        solutions.append(xi)
        print(f"  x{i+1} = det(A{i+1}) / det(A) = ({sympy.simplify(det_Ai)}) / ({sympy.simplify(det_A)}) = {xi}")

    print("\nConclusión (c): La solución única para un α dado (distinto de los valores críticos) es:")
    for i, sol in enumerate(solutions):
        print(f"  x{i+1} = {sol}")

alpha = sympy.Symbol('α')

A_system = sympy.Matrix([
    [1,     -1,    alpha],
    [-1,     2,   -alpha],
    [alpha,  1,        1]
])

b_system = sympy.Matrix([-2, 3, 2])

analyze_system_with_parameter(A_symbolic=A_system, b_symbolic=b_system, symbol=alpha)
```

```
--- Análisis del Determinante ---
La matriz de coeficientes A es:
Matrix([[1, -1, α], [-1, 2, -α], [α, 1, 1]])

El determinante de A es: det(A) = 1 - α**2

Se busca para qué valores del símbolo el determinante es cero.
El determinante es cero cuando α está en [-1, 1].
------------------------------------------------------------
--- Análisis de los Casos Críticos (No hay solución única) ---

Análisis para α = -1:
Matriz aumentada para este valor:
 Matrix([[1, -1, -1, -2], [-1, 2, 1, 3], [-1, 1, 1, 2]])

Forma escalonada reducida por filas (RREF):
 Matrix([[1, 0, -1, -1], [0, 1, 0, 1], [0, 0, 0, 0]])

Conclusión (b): Para α = -1, el sistema es consistente y tiene INFINITAS SOLUCIONES.

Análisis para α = 1:
Matriz aumentada para este valor:
 Matrix([[1, -1, 1, -2], [-1, 2, -1, 3], [1, 1, 1, 2]])

Forma escalonada reducida por filas (RREF):
 Matrix([[1, 0, 1, 0], [0, 1, 0, 0], [0, 0, 0, 1]])

Conclusión (a): Para α = 1, el sistema es inconsistente y NO TIENE SOLUCIONES.
------------------------------------------------------------
--- Análisis del Caso General (Solución única) ---
Existe una solución única siempre que α no sea uno de los valores críticos [-1, 1].

Usando la Regla de Cramer para encontrar la solución en términos del símbolo:
  x1 = det(A1) / det(A) = (-α - 1) / (1 - α**2) = 1/(α - 1)
  x2 = det(A2) / det(A) = (1 - α**2) / (1 - α**2) = 1
  x3 = det(A3) / det(A) = (α + 1) / (1 - α**2) = -1/(α - 1)

Conclusión (c): La solución única para un α dado (distinto de los valores críticos) es:
  x1 = 1/(α - 1)
  x2 = 1
  x3 = -1/(α - 1)
```

7) Repita el ejercicio 4 con el método Gauss-Jordan.

In [ ]:
```
import numpy as np
import logging
from sys import stdout
from datetime import datetime

logging.basicConfig(
    level=logging.INFO,
    format="[%(asctime)s][%(levelname)s] %(message)s",
    stream=stdout,
    datefmt="%m-%d %H:%M:%S",
)
logging.info(datetime.now())

def gauss_jordan(A: np.ndarray | list[list[float | int]]) -> np.ndarray:
    """
    Resuelve un sistema de ecuaciones lineales mediante el método de Gauss-Jordan.

    Parameters
    ----------
    A : matriz aumentada del sistema de ecuaciones lineales (n x n+1)

    Returns
    -------
    solucion : vector con la solución del sistema de ecuaciones lineales
    """
    if not isinstance(A, np.ndarray):
        logging.debug("Convirtiendo A a numpy array.")
        A = np.array(A, dtype=np.float32)
    assert A.shape[0] == A.shape[1] - 1, "La matriz A debe ser de tamaño n-by-(n+1)."
    n = A.shape[0]

    for i in range(n):
        # Pivoteo parcial si es necesario
        max_row = np.argmax(np.abs(A[i:, i])) + i
        if A[max_row, i] == 0:
            raise ValueError("No existe solución única.")
        if max_row != i:
            A[[i, max_row]] = A[[max_row, i]]

        A[i] = A[i] / A[i, i]

        for j in range(n):
            if j != i:
                A[j] = A[j] - A[j, i] * A[i]

        logging.info(f"\n{A}")

    solucion = A[:, -1]
    return solucion
```

```
[01-07 21:51:38][INFO] 2026-01-07 21:51:38.991521
```

a.

In [72]:
```
A_aaa = [
    [1/4,1/5,1/6,9],
    [1/3,1/4,1/5,8],
    [1/2,1,2,8]
]

gauss_jordan(A_aaa)
```

```
[01-07 21:51:39][INFO]
[[ 1.         2.         4.        16.       ]
 [ 0.        -0.4166667 -1.1333333  2.6666665]
 [ 0.        -0.3       -0.8333333  5.       ]]
[01-07 21:51:39][INFO]
[[ 1.0000000e+00  0.0000000e+00 -1.4399996e+00  2.8799999e+01]
 [-0.0000000e+00  1.0000000e+00  2.7199998e+00 -6.3999991e+00]
 [ 0.0000000e+00  0.0000000e+00 -1.7333329e-02  3.0800002e+00]]
[01-07 21:51:39][INFO]
[[   1.         0.         0.      -227.07693]
 [   0.         1.         0.       476.92322]
 [  -0.        -0.         1.      -177.69237]]
```

Out[72]:
```
array([-227.07693,  476.92322, -177.69237], dtype=float32)
```

b.

In [73]:
```
A_abb = [
    [3.333,15920,-10.333,15913],
    [2.222,16.71,9.612,28.544],
    [1.5611,5.1791,1.6852,8.4254]
]

gauss_jordan(A_abb)
```

```
[01-07 21:51:39][INFO]
[[ 1.0000000e+00  4.7764775e+03 -3.1002102e+00  4.7743774e+03]
 [ 0.0000000e+00 -1.0596623e+04  1.6500668e+01 -1.0580122e+04]
 [ 0.0000000e+00 -7.4513799e+03  6.5249386e+00 -7.4448555e+03]]
[01-07 21:51:39][INFO]
[[ 1.0000000e+00  0.0000000e+00  4.3375430e+00  5.3378906e+00]
 [-0.0000000e+00  1.0000000e+00 -1.5571628e-03  9.9844283e-01]
 [ 0.0000000e+00  0.0000000e+00 -5.0780735e+00 -5.0786133e+00]]
[01-07 21:51:39][INFO]
[[ 1.         0.         0.         0.9998865]
 [-0.         1.         0.         1.0000001]
 [-0.        -0.         1.         1.0001063]]
```

Out[73]:
```
array([0.9998865, 1.0000001, 1.0001063], dtype=float32)
```

c.

In [74]:
```
A_acc = [
    [1,1/2,1/3,1/4,1/6],
    [1/2,1/3,1/4,1/5,1/7],
    [1/3,1/4,1/5,1/6,1/8],
    [1/4,1/5,1/6,1/7,1/9]
]

gauss_jordan(A_acc)
```

```
[01-07 21:51:39][INFO]
[[1.         0.5        0.33333334 0.25       0.16666667]
 [0.         0.08333334 0.08333333 0.075      0.05952381]
 [0.         0.08333333 0.08888888 0.08333334 0.06944444]
 [0.         0.075      0.08333334 0.08035715 0.06944445]]
[01-07 21:51:39][INFO]
[[ 1.          0.         -0.16666657 -0.19999996 -0.19047616]
 [ 0.          1.          0.9999998   0.8999999   0.7142857 ]
 [ 0.          0.          0.00555557  0.00833335  0.00992064]
 [ 0.          0.          0.00833335  0.01285715  0.01587302]]
[01-07 21:51:39][INFO]
[[ 1.0000000e+00  0.0000000e+00  0.0000000e+00  5.7142526e-02
   1.2698348e-01]
 [ 0.0000000e+00  1.0000000e+00  0.0000000e+00 -6.4285570e-01
  -1.1904728e+00]
 [ 0.0000000e+00  0.0000000e+00  1.0000000e+00  1.5428559e+00
   1.9047589e+00]
 [ 0.0000000e+00  0.0000000e+00  0.0000000e+00 -2.3809634e-04
  -6.6138338e-04]]
[01-07 21:51:39][INFO]
[[ 1.          0.          0.          0.         -0.03174686]
 [ 0.          1.          0.          0.          0.5952499 ]
 [ 0.          0.          1.          0.         -2.380982  ]
 [-0.         -0.         -0.          1.          2.7777972 ]]
```

Out[74]:
```
array([-0.03174686,  0.5952499 , -2.380982  ,  2.7777972 ], dtype=float32)
```

d.

In [75]:
```
A_add = [
    [2.,1,-1,1,-3,7],
    [1,0,2,-1,1,2],
    [0,-2,-1,1,-1,-5],
    [3,1,-4,0,5,6],
    [1,-1,-1,-1,1,-3]
]

gauss_jordan(A_add)
```

```
        [01-07 21:51:39][INFO]
        [[ 1.          0.33333334 -1.3333334   0.          1.6666666   2.        ]
         [ 0.         -0.33333334  3.3333335  -1.         -0.6666666   0.        ]
         [ 0.         -2.         -1.          1.         -1.         -5.        ]
         [ 0.          0.3333333   1.6666667   1.         -6.333333    3.        ]
         [ 0.         -1.3333334   0.33333337 -1.         -0.6666666  -5.        ]]
        [01-07 21:51:39][INFO]
        [[ 1.0000000e+00  0.0000000e+00 -1.5000000e+00  1.6666667e-01
           1.5000000e+00  1.1666666e+00]
         [-0.0000000e+00  1.0000000e+00  5.0000000e-01 -5.0000000e-01
           5.0000000e-01  2.5000000e+00]
         [ 0.0000000e+00  0.0000000e+00  3.5000002e+00 -1.1666666e+00
          -4.9999994e-01  8.3333337e-01]
         [ 0.0000000e+00  0.0000000e+00  1.5000001e+00  1.1666666e+00
          -6.4999995e+00  2.1666667e+00]
         [ 0.0000000e+00  0.0000000e+00  1.0000000e+00 -1.6666667e+00
           5.9604645e-08 -1.6666665e+00]]
        [01-07 21:51:39][INFO]
        [[ 1.          0.          0.         -0.3333333   1.2857144   1.5238094 ]
         [-0.          1.          0.         -0.33333334  0.57142854  2.3809524 ]
         [ 0.          0.          1.         -0.3333333  -0.14285712  0.23809524]
         [ 0.          0.          0.          1.6666666  -6.2857137   1.8095238 ]
         [ 0.          0.          0.         -1.3333335   0.14285718 -1.9047618 ]]
        [01-07 21:51:39][INFO]
        [[ 1.          0.          0.          0.          0.02857172  1.8857142 ]
         [ 0.          1.          0.          0.         -0.68571424  2.7428572 ]
         [ 0.          0.          1.          0.         -1.3999997   0.6       ]
         [ 0.          0.          0.          1.         -3.7714283   1.0857143 ]
         [ 0.          0.          0.          0.         -4.8857145  -0.45714247]]
        [01-07 21:51:39][INFO]
        [[ 1.          0.          0.          0.          0.          1.8830408 ]
         [ 0.          1.          0.          0.          0.          2.8070176 ]
         [ 0.          0.          1.          0.          0.          0.73099405]
         [ 0.          0.          0.          1.          0.          1.4385962 ]
         [-0.         -0.         -0.         -0.          1.          0.09356717]]
Out[75]: array([1.8830408 , 2.8070176 , 0.73099405, 1.4385962 , 0.09356717],
               dtype=float32)
```